

Tarun Kumar

ROS2 — Nav2 — Autonomous Robotics — UAV Systems

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SUMMARY

Robotics and Automation Engineering student specializing in ROS2-based autonomous systems, UAV development, and perception. Experienced in Nav2, SLAM, LiDAR integration, and AI-based vision.

EDUCATION

Sahyadri College of Engineering and Management
B.E. Robotics and Automation Engineering

2023 – Present
Mangaluru

TECHNICAL SKILLS

Robotics: ROS2, Nav2, SLAM Toolbox, Cartographer
UAV: ArduPilot, MAVLink
Embedded: Raspberry Pi, ESP32, Arduino
CAD: SOLIDWORKS (CSWA), Fusion 360

Programming: C++, Python, Bash
Computer Vision: YOLOv5, OpenCV
Simulation: Gazebo, MATLAB
Tools: ANSYS, LabVIEW

PROJECTS

Autonomous Mobile Robot

GitHub

ROS2, Nav2, SLAM Toolbox, LiDAR

- Built ROS2-based robot using Nav2 for autonomous navigation with path planning and obstacle avoidance
- Integrated LiDAR with SLAM Toolbox for real-time mapping and localization
- Developed ROS2 nodes, topics, and TF tree for full navigation pipeline

AgriDrone Audit System

GitHub

YOLOv5, OpenCV, UAV Systems

- Developed UAV-based crop monitoring system using YOLOv5 achieving ~85% accuracy
- Implemented real-time onboard image processing for aerial analytics
- Designed scalable system architecture for agricultural monitoring

OPEN SOURCE CONTRIBUTIONS

- Contributed to ROS2 Nav2 by improving Costmap2D documentation and parameter clarity; PR merged by maintainer

EXPERIENCE

Project Intern — NIT Karnataka

Feb 2025 – May 2025

Robotics Research

Mangaluru

- Worked on autonomous robotics system integration and ROS2-based data pipelines

Team Member — Team Challengers

Oct 2023 – Feb 2025

Robotics Team

Mangaluru

- Developed UAV and robotics hardware systems and participated in testing

Design Assistant Intern — Atharva Enterprise

Jun 2024 – Dec 2024

CAD Design

Mangaluru

- Designed mechanical components using SOLIDWORKS

CERTIFICATIONS

- SOLIDWORKS CAD Design Associate (CSWA)
- MATLAB & Simulink Onramp
- Python Foundation Certification